$$\Delta\theta \sim \frac{\lambda}{2a}$$

Thus, when $a >>> \lambda$ the intensity at a point R (which is deep inside the geometrical shadow) will be negligible; on the other hand, if $a \sim \lambda$ there will be almost uniform spreading out of the beam resulting in an (almost) uniform illumina-tion of the screen. This phenomenon is a manifestation of the fact that when $a >>> \lambda$, the secondary wavelets emanating from different points on the circular aperture so beautifully interfere to produce (almost) zero intensity in the geometrical shadow and a large intensity inside the circular region (see Fig. 20.1). However, if $a \sim \lambda$ then the aperture almost acts as a point source resulting in a uniform illumination of the screen (see Fig. 12.4 and Fig. 22 in prelims pages).

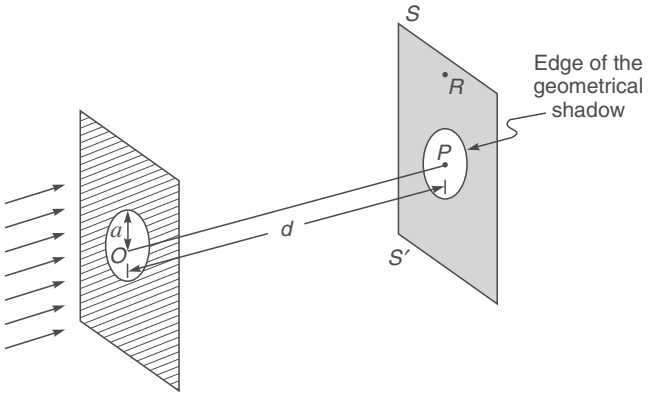


Fig. 20.1 Diffraction of a plane wave incident normally on a circular aperture of radius a.

We will first introduce the concept of Fresnel half-period zones to have a qualitative understanding of the Fresnel diffraction pattern; this will be followed by a more rigorous analysis of the Fresnel class of diffraction and its transition to the Fraunhofer region.

20.2 FRESNEL HALF-PERIOD ZONES

LO 1

Let us consider a plane wavefront WW′ propagating in the z-direction as shown in Fig. 20.2. In order to determine the field at an arbitrary point P due to the disturbances reaching from different portions of the wavefront, we make the following construction: from the point P we drop a perpendicular PO on the wavefront. If $PO = d$, then with point P as centre we draw spheres of radii $d + \lambda/2$, $d + 2\lambda/2$, $d + 3\lambda/2$,…; these spheres will intersect WW′ in circles as shown in Fig. 20.2. The radius of the nth circle will obviously be given by

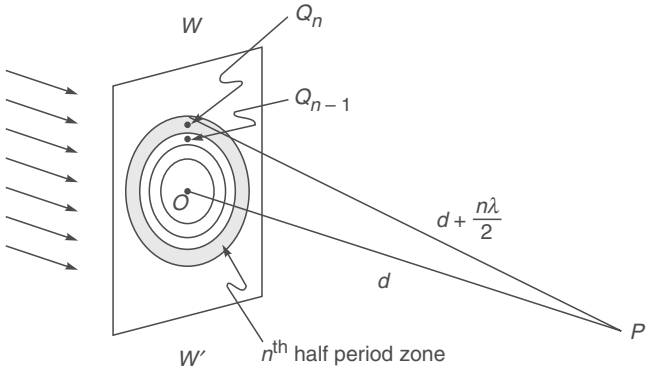


Fig. 20.2 Construction of Fresnel half-period zones.

$$r_n = \left[\left(d + n\frac{\lambda}{2}\right)^2 - d^2\right]^{1/2}$$

$$= \sqrt{n\lambda d}\left[1 + \frac{n\lambda}{4d}\right]^{1/2}$$

or

$$r_n \approx \sqrt{n\lambda d} \qquad \textbf{(20.1)}$$

where we have assumed $d >>> \lambda$, this is indeed justified for practical systems using visible light; of course, we are assuming that n is not a very large number. The annular region between the nth circle and $(n-1)^{th}$ circle is known as the nth **half-period zone**; the area of the nth half-period zone is given by

$$A_n = \pi r_n^2 - \pi r_{n-1}^2$$

$$\approx \pi[n\lambda d - (n-1)\lambda d] = \pi\lambda d \qquad \textbf{(20.2)}$$

Thus the areas of all the half-period zones are approximately equal. Now the resultant disturbance produced by the nth zone will be π out-of-phase with the disturbance produced by the $(n-1)^{th}$ [or the $(n+1)^{th}$] zone. This can easily be seen from the following consideration: For infinitesimal area surrounding a point Q_n in the nth half period zone, there is a corresponding infinitesimal area surrounding the point Q_{n-1} in the $(n-1)^{th}$ half-period zone such that

$$Q_nP - Q_{n-1}P = \frac{\lambda}{2}$$

which corresponds to a phase difference of π. Since the areas of the zones are approximately equal, one can have a one-to-one correspondence between points in various zones. Thus, the resultant amplitude at the point P can be written as

$$u(P) = u_1 - u_2 + u_3 - u_4 + \ldots + (-1)^{m+1} u_m + \ldots. \qquad \textbf{(20.3)}$$

where u_n represents the net amplitude produced by the secondary wavelets emanating from the nth zone; the alternate negative and positive signs represent the fact that the result-

ant disturbances produced by two consecutive zones are π out-of-phase with respect to each other. The amplitude produced by a particular zone is proportional to the area of the zone and inversely proportional to the distance of the zone from the point P; further, it also depends on an obliquity factor which is proportional to $\frac{1}{2}(1+\cos\chi)$ where χ is the angle that the normal to the zone makes with the line QP; this obliquity factor comes out automatically from rigorous diffraction theory.* Thus, we may write

$$u_n = \text{constant}\ \frac{A_n}{Q_n P}\frac{(1+\cos\chi)}{2} \tag{20.4}$$

where A_n represents the area of the nth zone. It can be shown that if we use the exact expression for r_n, the area of the zones increase with n; however, this slight increase in the area is exactly compensated by the increased distance of the zone from the point P. In spite of this, the amplitudes $u_1, u_2, u_3 \ldots$ decrease monotonically because of increased obliquity. Thus, we may write

$$u_1 > u_2 > u_3 \ldots \tag{20.5}$$

The series expressed by Eq. (20.3) can be approximately summed due to a method by Schuster. We rewrite Eq. (20.3) as

$$u(P) = \frac{u_1}{2} + \left(\frac{u_1}{2} - u_2 + \frac{u_3}{2}\right) + \left(\frac{u_3}{2} - u_4 + \frac{u_5}{2}\right) + \ldots \tag{20.6}$$

where the last term would either $\frac{1}{2}u_m$ or $\left(\frac{1}{2}u_{m-1} - u_m\right)$ according to m being odd or even. If the obliquity factor is such that

$$u_n > \frac{1}{2}(u_{n-1} + u_{n+1}) \tag{20.7}$$

then the quantities inside the brackets in Eq. (20.6) will be negative; consequently,

$$\left.\begin{aligned} &u(P) < \frac{1}{2}u_1 + \frac{1}{2}u_m \ ;(m \text{ odd}) \\ \text{and}\quad &u(P) < \frac{1}{2}u_1 + \frac{1}{2}u_{m-1} - u_m \approx \frac{u_1}{2} - \frac{u_m}{2} \ ;(m \text{ even}) \end{aligned}\right\} \tag{20.8}$$

where we have assumed that the amplitude of the fields produced by consecutive zones differ only slightly. In order to obtain the upper limits, we rewrite Eq. (20.3) in the form

$$u(P) = u_1 - \frac{u_2}{2} - \left(\frac{u_2}{2} - u_3 + \frac{u_4}{2}\right) - \left(\frac{u_4}{2} - u_5 + \frac{u_6}{2}\right) \tag{20.9}$$

where the last term would now be $\left(-\frac{1}{2}u_{m-1} + u_m\right)$ when m is odd and $-\frac{1}{2}u_m$ when m is even. Since the quantities inside the brackets are negative, we obtain

$$\left.\begin{aligned} &u(P) > u_1 - \frac{u_2}{2} - \frac{u_{m-1}}{2} + u_m \approx \frac{u_1}{2} + \frac{u_m}{2} \\ &\qquad\qquad (\text{when } m \text{ is odd}) \\ \text{and}\quad &u(P) > u_1 - \frac{u_2}{2} - \frac{u_m}{2} \approx \frac{u_1}{2} - \frac{u_m}{2} \\ &\qquad\qquad (\text{when } m \text{ is even}) \end{aligned}\right\} \tag{20.10}$$

Using Eqs. (20.8) and (20.10) we may approximately write

$$\left.\begin{aligned} &u(P) \approx \frac{u_1}{2} + \frac{u_m}{2}\ \ (\text{when } m \text{ is odd}) \\ \text{and}\quad &u(P) \approx \frac{u_1}{2} - \frac{u_m}{2}\ \ (\text{when } m \text{ is even}) \end{aligned}\right\} \tag{20.11}$$

If we can neglect u_m in comparison to u_1 then the Eq. (20.11)** gives the remarkable result that

$$u(P) \approx \frac{u_1}{2} \tag{20.12}$$

implying that *the resultant amplitude produced by the entire wavefront is only one half of the amplitude produced by the first half-period zone.*

20.2.1 Diffraction by a Circular Aperture

We may use the above analysis to study the diffraction of a plane wave by a circular aperture. Let the point P be at a distance d from the circular aperture (see Fig. 20.1). We assume that the radius of the circular aperture, a, can be increased from zero onwards. As a increases, the intensity at the point P would also increase till the circular aperture contains the first half-period zone; this would happen when $a = \sqrt{\lambda d}$. The resultant amplitude at the point P would be u_1 which is twice the value of the amplitude for the unobstructed wavefront [see Eq. (20.12)]. The intensity would therefore be $4I_0$, where I_0 represents the intensity at the point P due to the unobstructed wavefront. If we further increase

* See, e.g., Ref. 20.3.

** If one assumes a form of the obliquity factor as given by Eq. (20.4) then it decreases from 1 to 1/2 as m increases from 1 to ∞; this implies that $|u_m|$ can never be smaller that $\frac{u_1}{2}$. However, when m is large, a slight shift of the point P on the axis will change the amplitude from $\frac{u_1}{2} + \frac{u_m}{2}$ to $\frac{u_1}{2} - \frac{u_m}{2}$, the changes will occur with such great rapidity that one can only observe the average value which will be $\frac{u_1}{2}$.

a then $u(P)$ would start decreasing and when the circular aperture contains the first two half-period zones (which would happen when $a = \sqrt{2\lambda d}$) the resultant amplitude ($= u_1 - u_2$) would be almost zero. *Thus, by increasing the hole diameter, the intensity at the point P decreases almost to zero*. This interesting result is once again due to the validity of the Huygens–Fresnel principle and hence would be valid for sound waves also. We may generalize the above result by noting that if

$$a = \sqrt{(2n+1)\lambda d}\ ; n = 0, 1, 2,\ldots \text{ [maxima]}$$

the aperture will contain an odd number of half-period zones and the intensity will be maximum; on the other hand, if

$$a = \sqrt{2n\lambda d}\ ; n = 1, 2,\ldots \text{ [minima]}$$

the aperture will contain an even number of half-period zones and the intensity will be minimum. In order to have a numerical appreciation, we note that for $d = 50$ cm and $\lambda = 5 \times 10^{-5}$ cm, the radii of the first, second and third zones would be 0.500 mm, 0.707 mm and 0.866 mm, respectively. As a corollary of the above analysis we can consider a circular aperture of a fixed radius a and study the intensity variation along the axis. Whenever the distance

$$d = \frac{a^2}{(2n+1)\lambda}\ ; n = 0, 1, 2,\ldots \text{ (maxima)}$$

the point P (see Fig. 20.1) will correspond to a maximum. Similarly, when

$$d = \frac{a^2}{2n\lambda}\ ; n = 1, 2,\ldots \text{ (minima)}$$

the point P will correspond to a minimum. The intensity distribution on a screen SS' at off-axis points can be approximately calculated by using the half-period zones, but such a calculation is fairly cumbersome. However, from the symmetry of the problem, one can deduce that the diffraction pattern has to be in the form of concentric circular rings with their centres at the point P.

20.2.2 Diffraction by an Opaque Disc–The Poisson Spot

If instead of the circular aperture we have a circular disc [see Fig. 20.3(a)] and if the disc obstructs the first p half-period zones then the field at the point P would be

$$u(P) = u_{p+1} - u_{p+2} + \ldots \approx \frac{u_{p+1}}{2} \qquad \textbf{(20.13)}$$

Thus, we should always obtain a bright spot on the axis behind a circular disc; (the more rigorous theory also predicts the same result — see Sec. 20.4.2). This is called the '**Poisson spot**'. We may mention here that it was in 1816 that the French physicist Augustin Fresnel developed the mathematical theory of diffraction using the wave theory of light. Simeon Poisson, the famous mathematician, used Fresnel's theory to predict a bright spot at the center of the shadow of an opaque disc. Poisson was a great supporter of the corpuscular theory of light and he said that since the bright spot is against common sense, the wave theory must be wrong. Shortly afterwards, Fresnel and Arago carried out the experiment to demonstrate the existence of the Poisson spot [see Fig. 20.3(b)], validating the wave theory.

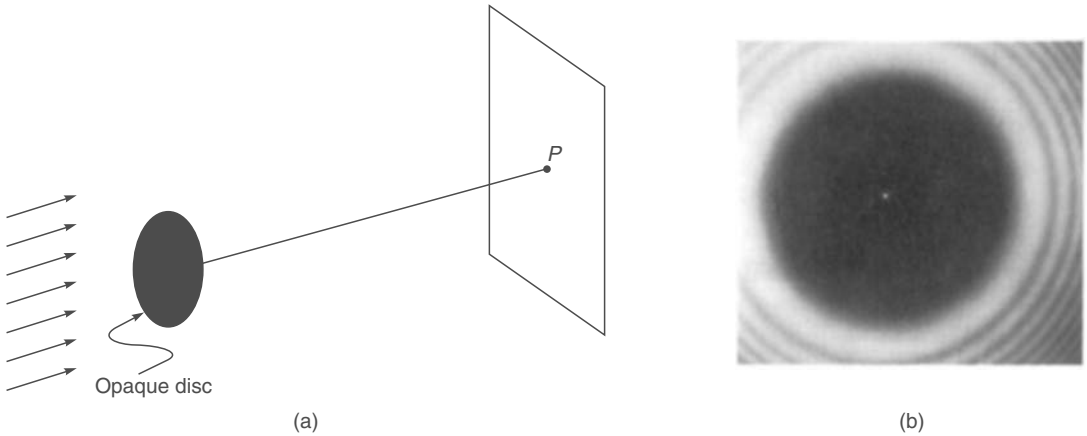


Fig. 20.3 (a) When a plane wave is incident normally on an opaque disc, a bright spot is always formed on an axial point. This spot is known as the Poisson spot. (b) The Poisson spot at the center of the shadow of a one cent coin; the screen is 20 m from the coin and the source of light is also 20 m from the coin. [Photograph reprinted with permission from P.M. Rinard, *Large scale diffraction patterns from circular objects*, American Journal of Physics, Vol 44, 70, 1976; Copyright (1976), American Association of Physics Teachers used with Permission].

20.3 THE ZONE-PLATE LO 1

A beautiful application of the concept of Fresnel half-period zones lies in the construction of the zone-plate which consists of a large number of concentric circles whose radii are proportional to the square root of natural numbers and the alternate annular regions of which are blackened (see Fig. 20.4). Let the radii of the circles be $\sqrt{1}\,K$, $\sqrt{2}\,K$, $\sqrt{3}\,K$, $\sqrt{4}\,K$,… where K is a constant and has the dimension of length. We consider a point P_1 which is at a distance K^2/λ from the zone plate; for this point the blackened rings correspond to the 2nd, 4th, 6th, … half-period zones. Thus, the even zones are obstructed and the resultant amplitude at P_1 [see Fig. 20.5 (a)] will be

$$u_1 + u_3 + u_5 + \ldots \qquad \textbf{(20.14)}$$

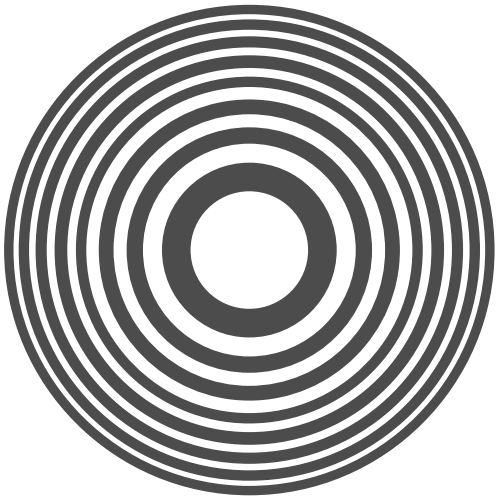


Fig. 20.4 The zone plate.

producing an intense maximum. For the point P_3 (which is at a distance $K^2/3\lambda$) the first blackened ring contains the 4th, 5th, 6th zones, the second blackened ring contains the 10th, 11th and 12th zones, etc.; thus the resultant amplitude would be

$$(u_1 - u_2 + u_3) + (u_7 - u_8 + u_9) + \ldots \qquad \textbf{(20.15)}$$

which would again correspond to a maximum, but it would not be as intense as the point P_1. Between the points P_1 and P_3 there will be a point P_2 (at a distance $K^2/2\lambda$) where the resultant amplitude would be

$$(u_1 - u_2) + (u_5 - u_6) + \ldots \qquad \textbf{(20.16)}$$

implying that corresponding to P_2 the first blackened ring contains the 3rd and 4th half-period zones, etc. Obviously, the point P_2 will correspond to a minimum. Thus, if a plane wave is incident normally on a zone-plate, then the corresponding focal points are at distances

$$\frac{K^2}{\lambda}, \frac{K^2}{3\lambda}, \frac{K^2}{5\lambda}, \ldots \qquad \textbf{(20.17)}$$

from the zone-plate. Elementary calculations will show that the zone-plate suffers from considerable chromatic aberrations [see Problem 20.5].

Example 20.1 Assume a plane wave ($\lambda = 5 \times 10^{-5}$ cm) to be incident on a circular aperture of radius 0.5 mm. We will calculate the positions of the brightest and darkest points on the axis. For the brightest point, the aperture should contain only the first zone and thus we must have (see Fig. 20.1)

$$(0.05)^2 = OP\,(5 \times 10^{-5})$$

Thus OP = 50 cm. Similarly, the darkest point would be at a distance

$$\frac{(0.05)^2}{2 \times 5 \times 10^{-5}} = 25 \text{ cm}$$

Example 20.2 Consider a zone-plate with radii

$$r_n = 0.1\sqrt{n} \text{ cm}$$

For $\lambda = 5 \times 10^{-5}$ cm, we will calculate the positions of various foci. The most intense focal point will be at a distance

$$\frac{r_1^2}{\lambda} = \frac{0.01}{5 \times 10^{-5}} = 200 \text{ cm}$$

The other focal points will be at distances of 200/3, 200/5, 200/7 cm, etc. Between any two consecutive foci there will be dark points on the axis corresponding to which the first circle will contain an even number of half-period zones.

The zone-plate can also be used for imaging points on the axis, e.g., if we have a point source at S then a bright image will be formed at P, where the point P should be such that (see Fig. 20.5(b)):

$$SL + LP - SP = \frac{\lambda}{2} \qquad \textbf{(20.18)}$$

the point L being on the periphery of the first circle of the zone plate [see Fig. 20.5(b)]. If the radius of the first circle is r_1, then

$$\begin{aligned} SL + LP - SP &= \sqrt{a^2 + r_1^2} + \sqrt{b^2 + r_1^2} - (a + b) \\ &\approx a\left[1 + \frac{r_1^2}{2a^2}\right] + b\left[1 + \frac{r_1^2}{2b^2}\right] - (a + b) \\ &\approx \frac{r_1^2}{2}\left(\frac{1}{a} + \frac{1}{b}\right) \end{aligned} \qquad \textbf{(20.19)}$$

Thus Eq. (20.18) becomes

$$\frac{1}{a} + \frac{1}{b} = \frac{1}{f} \qquad \textbf{(20.20)}$$

where $f = r_1^2/\lambda$ represents the focal length. Equation (20.20) resembles the lens law. A very interesting demonstration experiment of the zone-plate can be carried out by using